Enterprise Java Technology

Practical Assignment - II

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MCA_3

1. Create a JSP page to accept student details such as Student ID, Student Name, Course, Semester and Email, and display the submitted information on another JSP page.

student.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Details Form</title>
</head>
<body>
   <h2>Student Details Form</h2>

   <form action="display.jsp" method="post">
      Student ID: <input type="text" name="studentId" required><br><br>

      Student Name: <input type="text" name="studentName" value="Dhruv" required><br><br>

      Course: <input type="text" name="course" value="MCA_01" required><br><br>

      Semester: <input type="text" name="semester" value="01" required><br><br>

      Email: <input type="email" name="email" required><br><br>

      <input type="submit" value="Submit">
      <input type="reset" value="Reset">
   </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Details</title>
</head>
<body>
   <h2>Submitted Student Details</h2>

   <%
      String studentId = request.getParameter("studentId");
      String studentName = request.getParameter("studentName");
      String course = request.getParameter("course");
      String semester = request.getParameter("semester");
      String email = request.getParameter("email");
   %>

   <table border="1" cellpadding="10">
      <tr>
         <th>Student ID</th>
         <td><%= studentId %></td>
      </tr>
      <tr>
         <th>Student Name</th>
         <td><%= studentName %></td>
      </tr>
      <tr>
         <th>Course</th>
         <td><%= course %></td>
      </tr>
      <tr>
         <th>Semester</th>
         <td><%= semester %></td>
      </tr>
      <tr>
         <th>Email</th>
         <td><%= email %></td>
      </tr>
   </table>

   <br>
   <a href="student.jsp">Back to Form</a>
</body>
</html>
```

2. Create a JSP page that accepts five subject marks and calculates the total, percentage and grade.

marks.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Marks</title>
</head>
<body>
   <h2>Student Marks Calculator</h2>
   <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

   <form action="result.jsp" method="post">
      Subject 1 Marks: <input type="number" name="m1" min="0" max="100" required><br><br>
      Subject 2 Marks: <input type="number" name="m2" min="0" max="100" required><br><br>
      Subject 3 Marks: <input type="number" name="m3" min="0" max="100" required><br><br>
      Subject 4 Marks: <input type="number" name="m4" min="0" max="100" required><br><br>
      Subject 5 Marks: <input type="number" name="m5" min="0" max="100" required><br><br>

      <input type="submit" value="Calculate Result">
      <input type="reset" value="Reset">
   </form>
</body>
</html>
```

result.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>Student Result</title>
</head>
<body>
   <h2>Student Result</h2>

   <%
      int m1 = Integer.parseInt(request.getParameter("m1"));
      int m2 = Integer.parseInt(request.getParameter("m2"));
```

```
        int m3 = Integer.parseInt(request.getParameter("m3"));
        int m4 = Integer.parseInt(request.getParameter("m4"));
        int m5 = Integer.parseInt(request.getParameter("m5"));

        int total = m1 + m2 + m3 + m4 + m5;
        double percentage = total / 5.0;
        String grade;

        if (m1 < 35 || m2 < 35 || m3 < 35 || m4 < 35 || m5 < 35) {
            grade = "Fail";
        } else if (percentage >= 90) {
            grade = "A+";
        } else if (percentage >= 80) {
            grade = "A";
        } else if (percentage >= 70) {
            grade = "B";
        } else if (percentage >= 60) {
            grade = "C";
        } else if (percentage >= 50) {
            grade = "D";
        } else {
            grade = "Pass";
        }
    %>

    <p><b>Name:</b> Dhruv</p>
    <p><b>Roll No:</b> 01</p>
    <p><b>Course:</b> MCA_01</p>

    <table border="1" cellpadding="10">
        <tr><th>Subject</th><th>Marks</th></tr>
        <tr><td>Subject 1</td><td><%= m1 %></td></tr>
        <tr><td>Subject 2</td><td><%= m2 %></td></tr>
        <tr><td>Subject 3</td><td><%= m3 %></td></tr>
        <tr><td>Subject 4</td><td><%= m4 %></td></tr>
        <tr><td>Subject 5</td><td><%= m5 %></td></tr>
        <tr><th>Total</th><th><%= total %> / 500</th></tr>
        <tr><th>Percentage</th><th><%= String.format("%.2f", percentage) %>%</th></tr>
        <tr><th>Grade</th><th><%= grade %></th></tr>
    </table>

    <br>
    <a href="marks.jsp">Back to Marks Form</a>
</body>
</html>
```

4. Create a JSP page that demonstrates the use of JSP declaration, scriptlet and expression tags.

index.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
   <title>JSP Tags Demonstration</title>
</head>
<body>
   <h2>JSP Declaration, Scriptlet and Expression Tags</h2>

   <%!
      String name = "Dhruv";
      int rollNo = 1;

      int square(int n) {
         return n * n;
      }
   %>

   <%
      String course = "MCA_01";
      int semester = 3;
      int number = 5;
   %>

   <h3>Student Details</h3>
   <p>Name: <%= name %></p>
   <p>Roll No: <%= rollNo %></p>
   <p>Course: <%= course %></p>
   <p>Semester: <%= semester %></p>

   <h3>Expression Tag Output</h3>
   <p>Number: <%= number %></p>
   <p>Square of <%= number %> is <%= square(number) %></p>

</body>
</html>
```

5. Create a JSP application to demonstrate the use of the following implicit objects:

request
response
session
application
Out

index.jsp

```
<%@ page language="java" %>
<html>
<head>
   <title>JSP Implicit Objects</title>
</head>
<body>
   <h2>JSP Implicit Objects</h2>

   <form method="post">
      Enter Name:
      <input type="text" name="name">
      <input type="submit" value="Submit">
   </form>

   <%
      String name = request.getParameter("name");

      if (name != null) {
         session.setAttribute("name", name);
         application.setAttribute("course", "MCA_01");

         out.println("<h3>Student Details</h3>");
         out.println("Request: " + name + "<br>");
         out.println("Response: " + response.getContentType() + "<br>");
         out.println("Session: " + session.getAttribute("name") + "<br>");
         out.println("Application: " + application.getAttribute("course") + "<br>");
         out.println("Out: Welcome " + name + "<br>");
         out.println("Roll No: 01");
      }
   %>

</body>
</html>
```

6. Create a JSP application that accepts username and password and displays a personalized welcome message using the request object.

```
<%@ page language="java" %>
<html>
<head>
   <title>Welcome Page</title>
</head>
<body>
   <h2>Login Form</h2>

   <form method="post">
      Username: <input type="text" name="username" required><br><br>
      Password: <input type="password" name="password" required><br><br>
      <input type="submit" value="Login">
   </form>

   <%
      String username = request.getParameter("username");
      String password = request.getParameter("password");

      if (username != null && password != null) {
   %>
      <h3>Welcome, <%= username %>!</h3>
      <p>You have successfully submitted your login details.</p>
   <%
      }
   %>

</body>
</html>
```

7. Create a JSP application that stores the logged-in user's username in a session and displays it on multiple JSP pages.

login.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Login Page</h2>
```

```
    <form action="welcome.jsp" method="post">
        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <input type="submit" value="Login">
    </form>
</body>
</html>
```

welcome.jsp

```
<%@ page language="java" %>
<html>
<body>
    <%
        String username = request.getParameter("username");
        session.setAttribute("username", username);
    %>

    <h2>Welcome, <%= session.getAttribute("username") %>!</h2>

    <a href="profile.jsp">Go to Profile</a>
</body>
</html>
```

profile.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Profile Page</h2>

    <%
        String username = (String) session.getAttribute("username");

        if (username != null) {
    %>
        <h3>Logged-in User: <%= username %></h3>
    <%
        } else {
    %>
        <h3>Please login first!</h3>
        <a href="login.jsp">Login</a>
    <%
        }
    %>
```

```
    <br><br>
    <a href="welcome.jsp">Go to Welcome Page</a>
</body>
</html>
```

8. Create a JSP application using include to display a common header and footer on multiple JSP pages.

header.jsp

```
<h2>Welcome to MCA_01</h2>
<hr>
```

footer.jsp

```
<hr>
<p>Name: Dhruv | Roll No: 01</p>
<p>© 2026 MCA Student Portal</p>
```

index.jsp

```
<%@ page language="java" %>
<html>
<body>

    <%@ include file="header.jsp" %>

    <h3>Home Page</h3>
    <p>Welcome to the Home Page.</p>

    <%@ include file="footer.jsp" %>

</body>
</html>
```

about.jsp

```
<%@ page language="java" %>
<html>
<body>

    <%@ include file="header.jsp" %>
```

```
    <h3>About Page</h3>
    <p>This is the About Page.</p>

    <%@ include file="footer.jsp" %>

</body>
</html>
```

9. Create a JSP application using the JSP forward action to transfer a request from one JSP page to another.

index.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>JSP Forward Example</h2>

    <form action="forward.jsp" method="post">
        Enter Name:
        <input type="text" name="name" required>
        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

forward.jsp

```
<%@ page language="java" %>
<%
    request.setAttribute("course", "MCA_01");
%>

<jsp:forward page="welcome.jsp" />
```

welcome.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Welcome Page</h2>
```

```
    <%
        String name = request.getParameter("name");
        String course = (String) request.getAttribute("course");
    %>

    <h3>Welcome, <%= name %>!</h3>
    <p>Course: <%= course %></p>
    <p>Name: Dhruv</p>
    <p>Roll No: 01</p>
</body>
</html>
```

10. Create a JSP page that uses JavaBean properties to accept and display student information.

Student.java

```
package mybean;

public class Student {
    private int id;
    private String name;
    private String course;
    private String email;

    public Student() {
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
```

```
    public String getCourse() {
        return course;
    }

    public void setCourse(String course) {
        this.course = course;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }
}
```

student.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Student Information</h2>

    <form action="display.jsp" method="post">
        Student ID: <input type="number" name="id" required><br><br>
        Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
        Course: <input type="text" name="course" value="MCA_01" required><br><br>
        Email: <input type="email" name="email" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" %>
<jsp:useBean id="s" class="mybean.Student" scope="request" />
<jsp:setProperty name="s" property="*" />

<html>
<body>
    <h2>Student Details</h2>

    Student ID: <jsp:getProperty name="s" property="id" /><br><br>
```

```
    Student Name: <jsp:getProperty name="s" property="name" /><br><br>
    Course: <jsp:getProperty name="s" property="course" /><br><br>
    Email: <jsp:getProperty name="s" property="email" /><br><br>

    Roll No: 01
</body>
</html>
```

11. Develop a JSP-based Student Result System that accepts marks of five subjects and displays:
Total Marks
Percentage
Grade
Pass/Fail Status
Use separate JSP pages for input and result.

input.jsp

```
<%@ page language="java" %>
<html>
<head>
    <title>Student Result System</title>
</head>
<body>
    <h2>Student Result System</h2>
    <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

    <form action="result.jsp" method="post">
        Subject 1: <input type="number" name="m1" min="0" max="100" required><br><br>
        Subject 2: <input type="number" name="m2" min="0" max="100" required><br><br>
        Subject 3: <input type="number" name="m3" min="0" max="100" required><br><br>
        Subject 4: <input type="number" name="m4" min="0" max="100" required><br><br>
        Subject 5: <input type="number" name="m5" min="0" max="100" required><br><br>

        <input type="submit" value="Calculate Result">
        <input type="reset" value="Reset">
    </form>
</body>
</html>
```

result.jsp

```
<%@ page language="java" %>
<html>
<head>
   <title>Student Result</title>
</head>
<body>
   <h2>Student Result</h2>

   <%
      int m1 = Integer.parseInt(request.getParameter("m1"));
      int m2 = Integer.parseInt(request.getParameter("m2"));
      int m3 = Integer.parseInt(request.getParameter("m3"));
      int m4 = Integer.parseInt(request.getParameter("m4"));
      int m5 = Integer.parseInt(request.getParameter("m5"));

      int total = m1 + m2 + m3 + m4 + m5;
      double percentage = total / 5.0;

      String grade;
      String status;

      if (m1 < 35 || m2 < 35 || m3 < 35 || m4 < 35 || m5 < 35) {
         grade = "F";
         status = "Fail";
      } else {
         status = "Pass";

         if (percentage >= 90)
            grade = "A+";
         else if (percentage >= 80)
            grade = "A";
         else if (percentage >= 70)
            grade = "B";
         else if (percentage >= 60)
            grade = "C";
         else if (percentage >= 50)
            grade = "D";
         else
            grade = "E";
      }
   %>

   <p>Name: Dhruv</p>
   <p>Roll No: 01</p>
   <p>Course: MCA_01</p>

   <table border="1" cellpadding="10">
      <tr>
```

```
            <th>Subject</th>
            <th>Marks</th>
        </tr>
        <tr><td>Subject 1</td><td><%= m1 %></td></tr>
        <tr><td>Subject 2</td><td><%= m2 %></td></tr>
        <tr><td>Subject 3</td><td><%= m3 %></td></tr>
        <tr><td>Subject 4</td><td><%= m4 %></td></tr>
        <tr><td>Subject 5</td><td><%= m5 %></td></tr>
        <tr><th>Total Marks</th><td><%= total %> / 500</td></tr>
        <tr><th>Percentage</th><td><%= String.format("%.2f", percentage) %>%</td></tr>
        <tr><th>Grade</th><td><%= grade %></td></tr>
        <tr><th>Status</th><td><%= status %></td></tr>
    </table>

    <br>
    <a href="input.jsp">Back to Input Page</a>
</body>
</html>
```

12. Develop a JSP-based Login System using session management. After successful login, display a dashboard containing the username and provide a logout facility.

login.jsp

```
<%@ page language="java" %>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
    <h2>Student Login</h2>

    <form action="validate.jsp" method="post">
        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <input type="submit" value="Login">
    </form>
</body>
</html>
```

validate.jsp

```
<%@ page language="java" %>
```

```
<%
   String username = request.getParameter("username");
   String password = request.getParameter("password");

   if ("Dhruv".equals(username) && "1234".equals(password)) {
      session.setAttribute("username", username);
      response.sendRedirect("dashboard.jsp");
   } else {
      response.sendRedirect("login.jsp");
   }
%>
```

dashboard.jsp

```
<%@ page language="java" %>
<%
   String username = (String) session.getAttribute("username");

   if (username == null) {
      response.sendRedirect("login.jsp");
      return;
   }
%>
<html>
<head>
   <title>Dashboard</title>
</head>
<body>
   <h2>Student Dashboard</h2>

   <h3>Welcome, <%= username %>!</h3>
   <p>Name: Dhruv</p>
   <p>Roll No: 01</p>
   <p>Course: MCA_01</p>

   <a href="logout.jsp">Logout</a>
</body>
</html>
```

logout.jsp

```
<%@ page language="java" %>
<%
   session.invalidate();
   response.sendRedirect("login.jsp");
%>
```

13. Create a JSP application to demonstrate session tracking using:
Cookies
HttpSession
URL Rewriting

index.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Session Tracking</h2>

   <form action="tracking.jsp" method="post">
      Enter Name:
      <input type="text" name="username" required>
      <input type="submit" value="Submit">
   </form>
</body>
</html>
```

tracking.jsp

```
<%@ page language="java" %>
<%
   String username = request.getParameter("username");

   if (username != null) {
      Cookie c = new Cookie("username", username);
      c.setMaxAge(60 * 60);
      response.addCookie(c);

      session.setAttribute("username", username);
   } else {
      username = (String) session.getAttribute("username");
   }

   if (username == null) {
      response.sendRedirect("index.jsp");
      return;
   }

   String url = response.encodeURL("welcome.jsp");
```

```
%>

<html>
<body>
   <h2>Session Tracking Methods</h2>

   <p>Cookie: <%= username %></p>
   <p>HttpSession: <%= session.getAttribute("username") %></p>

   <a href="<%= url %>">Continue using URL Rewriting</a>
</body>
</html>
```

welcome.jsp

```
<%@ page language="java" %>
<%
   String username = (String) session.getAttribute("username");

   if (username == null) {
      response.sendRedirect("index.jsp");
      return;
   }
%>

<html>
<body>
   <h2>Welcome Page</h2>

   <p>Welcome, <%= username %>!</p>
   <p>URL Rewriting is demonstrated through the encoded URL.</p>

   <a href="index.jsp">Back to Home</a>
</body>
</html>
```

14. Develop a JSP application that retrieves student records from a database using JDBC and displays them in an HTML table.

student.jsp

```
<%@ page import="java.sql.*" %>
<%@ page language="java" %>
```

```
<html>
<head>
  <title>Student Records</title>
</head>
<body>
  <h2>Student Records</h2>
  <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

  <table border="1" cellpadding="10">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Course</th>
      <th>Semester</th>
      <th>Email</th>
    </tr>

    <%
      try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        Connection con = DriverManager.getConnection(
          "jdbc:mysql://localhost:3306/javadb",
          "root",
          "1234"
        );

        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT * FROM student");

        while (rs.next()) {
    %>
    <tr>
      <td><%= rs.getInt("id") %></td>
      <td><%= rs.getString("name") %></td>
      <td><%= rs.getString("course") %></td>
      <td><%= rs.getInt("semester") %></td>
      <td><%= rs.getString("email") %></td>
    </tr>
    <%
        }
        con.close();
      } catch (Exception e) {
        out.println("<tr><td colspan='5'>" + e.getMessage() + "</td></tr>");
      }
    %>
  </table>
```

```
</body>
</html>
```

15. Create a JSP application to perform CRUD operations on an Employee table using JSP and JDBC.

employee,.jsp

```
<%@ page import="java.sql.*" %>
<%@ page language="java" %>
<html>
<head>
   <title>Employee CRUD</title>
</head>
<body>
   <h2>Employee Management System</h2>
   <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

   <form action="employee.jsp" method="post">
      Employee ID: <input type="number" name="id" required><br><br>
      Employee Name: <input type="text" name="name"><br><br>
      Department: <input type="text" name="department"><br><br>
      Salary: <input type="number" step="0.01" name="salary"><br><br>

      <button type="submit" name="action" value="insert">Insert</button>
      <button type="submit" name="action" value="update">Update</button>
      <button type="submit" name="action" value="delete">Delete</button>
   </form>

   <%
      Connection con = null;

      try {
         Class.forName("com.mysql.cj.jdbc.Driver");

         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/javadb",
            "root",
            "1234"
         );

         String action = request.getParameter("action");

         if (action != null) {
            int id = Integer.parseInt(request.getParameter("id"));
```

```
        if (action.equals("insert")) {
            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO employee VALUES (?, ?, ?, ?)"
            );
            ps.setInt(1, id);
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("department"));
            ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));
            ps.executeUpdate();
            ps.close();
            out.println("<p>Employee inserted successfully!</p>");
        }

        else if (action.equals("update")) {
            PreparedStatement ps = con.prepareStatement(
                "UPDATE employee SET name=?, department=?, salary=? WHERE id=?"
            );
            ps.setString(1, request.getParameter("name"));
            ps.setString(2, request.getParameter("department"));
            ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
            ps.setInt(4, id);

            int rows = ps.executeUpdate();
            ps.close();
            out.println(rows > 0 ? "<p>Employee updated successfully!</p>"
                                 : "<p>Employee not found!</p>");
        }

        else if (action.equals("delete")) {
            PreparedStatement ps = con.prepareStatement(
                "DELETE FROM employee WHERE id=?"
            );
            ps.setInt(1, id);

            int rows = ps.executeUpdate();
            ps.close();
            out.println(rows > 0 ? "<p>Employee deleted successfully!</p>"
                                 : "<p>Employee not found!</p>");
        }
    }
%>

<h3>Employee Records</h3>

<table border="1" cellpadding="10">
  <tr>
    <th>ID</th>
```

```
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
        </tr>

        <%
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM employee");

            while (rs.next()) {
        %>
        <tr>
            <td><%= rs.getInt("id") %></td>
            <td><%= rs.getString("name") %></td>
            <td><%= rs.getString("department") %></td>
            <td><%= rs.getDouble("salary") %></td>
        </tr>
        <%
            }
            rs.close();
            st.close();
        } catch (Exception e) {
            out.println("<p>Error: " + e.getMessage() + "</p>");
        } finally {
            if (con != null) {
                try { con.close(); } catch (SQLException e) { }
            }
        }
        %>
    </table>
</body>
</html>
```

16. Create a JSP application that displays employee details based on a department selected from a dropdown list.

employee.jsp

```
<%@ page import="java.sql.*" %>
<%@ page language="java" %>

<html>
<head>
    <title>Employee Details</title>
</head>
```

```
<body>
    <h2>Employee Details by Department</h2>
    <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

    <form method="post">
        Select Department:
        <select name="department">
            <option value="IT">IT</option>
            <option value="HR">HR</option>
            <option value="Finance">Finance</option>
        </select>
        <input type="submit" value="Search">
    </form>

    <br>

    <%
        String department = request.getParameter("department");

        if (department != null) {
            try {
                Class.forName("com.mysql.cj.jdbc.Driver");

                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/javadb",
                    "root",
                    "1234"
                );

                PreparedStatement ps = con.prepareStatement(
                    "SELECT * FROM employee WHERE department = ?"
                );
                ps.setString(1, department);

                ResultSet rs = ps.executeQuery();
    %>

    <h3>Department: <%= department %></h3>

    <table border="1" cellpadding="10">
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
        </tr>

        <%
```

```
            boolean found = false;

            while (rs.next()) {
                found = true;
        %>
        <tr>
            <td><%= rs.getInt("id") %></td>
            <td><%= rs.getString("name") %></td>
            <td><%= rs.getString("department") %></td>
            <td><%= rs.getDouble("salary") %></td>
        </tr>
        <%
            }

            if (!found) {
        %>
        <tr>
            <td colspan="4">No employees found.</td>
        </tr>
        <%
            }

            rs.close();
            ps.close();
            con.close();

            } catch (Exception e) {
                out.println("Error: " + e.getMessage());
            }
        }
    %>
    </table>

</body>
</html>
```

17. Create a JSP page that uses Expression Language (EL) to display request parameters without using JSP scriptlets.

index.jsp

```
<%@ page language="java" %>
<html>
<head>
```

```
    <title>Expression Language Example</title>
</head>
<body>
    <h2>Student Details Form</h2>

    <form action="display.jsp" method="post">
        Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
        Roll No: <input type="text" name="rollno" value="01" required><br><br>
        Course: <input type="text" name="course" value="MCA_01" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" %>
<html>
<head>
    <title>Student Details</title>
</head>
<body>
    <h2>Student Details Using EL</h2>

    <p>Student Name: ${param.name}</p>
    <p>Roll No: ${param.rollno}</p>
    <p>Course: ${param.course}</p>

</body>
</html>
```

18. Create a JSP application using EL to display values stored in: Request scope Session scope Application scope

index.jsp

```
<%@ page language="java" %>
<%
    request.setAttribute("rname", "Dhruv");
    session.setAttribute("srollno", "01");
    application.setAttribute("course", "MCA_01");
```

```
%>

<html>
<body>
   <h2>EL Scope Example</h2>

   Request Scope: ${requestScope.rname}<br><br>
   Session Scope: ${sessionScope.srollno}<br><br>
   Application Scope: ${applicationScope.course}<br><br>
</body>
</html>
```

19. Create a JSP page that demonstrates EL operators for:
Arithmetic
Relational
Logical
Conditional

operator.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>EL Operators Demonstration</h2>

   <h3>Arithmetic Operators</h3>
   Addition: ${10 + 5}<br>
   Subtraction: ${10 - 5}<br>
   Multiplication: ${10 * 5}<br>
   Division: ${10 / 5}<br><br>

   <h3>Relational Operators</h3>
   Equal: ${10 == 10}<br>
   Greater Than: ${10 > 5}<br>
   Less Than: ${5 < 10}<br><br>

   <h3>Logical Operators</h3>
   AND: ${true && true}<br>
   OR: ${true || false}<br>
   NOT: ${not false}<br><br>

   <h3>Conditional Operator</h3>
   Result: ${10 >= 5 ? "Pass" : "Fail"}<br><br>
```

```
</body>
</html>
```

20. Create a JSP application that uses EL to access properties of a JavaBean containing:
Student ID
Student Name
Course
Semester

Student.java

```
package mybean;

public class Student {
   private int id;
   private String name;
   private String course;
   private int semester;

   public Student() {
   }

   public int getId() {
      return id;
   }

   public void setId(int id) {
      this.id = id;
   }

   public String getName() {
      return name;
   }

   public void setName(String name) {
      this.name = name;
   }

   public String getCourse() {
      return course;
   }

   public void setCourse(String course) {
```

```
        this.course = course;
    }

    public int getSemester() {
        return semester;
    }

    public void setSemester(int semester) {
        this.semester = semester;
    }
}
```

student.jsp

```
<%@ page language="java" %>
<html>
<body>
    <h2>Student Information</h2>

    <form action="display.jsp" method="post">
        Student ID: <input type="number" name="id" required><br><br>
        Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
        Course: <input type="text" name="course" value="MCA_01" required><br><br>
        Semester: <input type="number" name="semester" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page language="java" %>
<jsp:useBean id="s" class="mybean.Student" scope="request" />
<jsp:setProperty name="s" property="*" />

<html>
<body>
    <h2>Student Details Using EL</h2>

    Student ID: ${s.id}<br><br>
    Student Name: ${s.name}<br><br>
    Course: ${s.course}<br><br>
    Semester: ${s.semester}<br><br>
</body>
</html>
```

21. Create a JSP page using EL to display values from a Java List and Map.

index.jsp

```
<%@ page import="java.util.*" %>
<%@ page language="java" %>

<%
   List<String> subjects = new ArrayList<>();
   subjects.add("Java");
   subjects.add("Python");
   subjects.add("JSP");

   Map<String, String> student = new HashMap<>();
   student.put("name", "Dhruv");
   student.put("rollno", "01");
   student.put("course", "MCA_01");

   request.setAttribute("subjects", subjects);
   request.setAttribute("student", student);
%>

<html>
<body>
   <h2>EL List and Map Example</h2>

   <h3>Student Details</h3>
   Name: ${student.name}<br>
   Roll No: ${student.rollno}<br>
   Course: ${student.course}<br>

   <h3>Subjects</h3>
   ${subjects[0]}<br>
   ${subjects[1]}<br>
   ${subjects[2]}<br>
</body>
</html>
```

22. Create a JSP application that uses EL to check whether a user is logged in and conditionally displays the user's dashboard or login message.

login.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Login</h2>

   <form action="dashboard.jsp" method="post">
      Username: <input type="text" name="username" required><br><br>
      <input type="submit" value="Login">
   </form>
</body>
</html>
```

dashboard.jsp

```
<%@ page language="java" %>

<%
   String username = request.getParameter("username");

   if (username != null && !username.trim().isEmpty()) {
      session.setAttribute("username", username);
   }
%>

<html>
<body>
   <h2>Student Dashboard</h2>

   <c:if test="${not empty sessionScope.username}">
      Welcome, ${sessionScope.username}!<br>
      Roll No: 01<br>
      Course: MCA_01
   </c:if>

   <c:if test="${empty sessionScope.username}">
      Please login first.
      <a href="login.jsp">Login</a>
   </c:if>
</body>
</html>
```

23. Develop a JSP application that accepts two numbers and performs arithmetic operations using EL instead of scriptlets.

index.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Arithmetic Calculator</h2>

   <form action="result.jsp" method="post">
      First Number: <input type="number" name="a" required><br><br>
      Second Number: <input type="number" name="b" required><br><br>
      <input type="submit" value="Calculate">
   </form>
</body>
</html>
```

result.jsp

```
<%@ page language="java" %>
<html>
<body>
   <h2>Arithmetic Results</h2>

   <p>Addition: ${param.a + param.b}</p>
   <p>Subtraction: ${param.a - param.b}</p>
   <p>Multiplication: ${param.a * param.b}</p>
   <p>Division: ${param.a / param.b}</p>
   <p>Remainder: ${param.a % param.b}</p>
</body>
</html>
```

24. Create a JSP application using JSTL <c:if> to determine whether a student has passed or failed based on percentage.

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<body>
   <h2>Student Result</h2>

   <form method="post">
      Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
      Roll No: <input type="text" name="rollno" value="01" required><br><br>
      Percentage: <input type="number" name="percentage"
                  min="0" max="100" step="0.01" required><br><br>
      <input type="submit" value="Check Result">
   </form>

   <c:if test="${not empty param.percentage}">
      <h3>Name: <c:out value="${param.name}"/></h3>
      <p>Roll No: <c:out value="${param.rollno}"/></p>
      <p>Percentage: <c:out value="${param.percentage}"/>%</p>

      <c:if test="${param.percentage >= 35}">
         <h3>Result: Pass</h3>
      </c:if>

      <c:if test="${param.percentage < 35}">
         <h3>Result: Fail</h3>
```

```
    </c:if>
  </c:if>
</body>
</html>
```

25. Create a JSP application using JSTL <c:choose>, <c:when> and <c:otherwise> to display grades based on student marks.

grade.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<body>
  <h2>Student Grade System</h2>

  <form method="post">
    Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
    Marks: <input type="number" name="marks" min="0" max="100" required><br><br>
    <input type="submit" value="Check Grade">
  </form>

  <c:if test="${not empty param.marks}">
    <p>Name: <c:out value="${param.name}"/></p>
    <p>Marks: <c:out value="${param.marks}"/></p>

    <c:choose>
      <c:when test="${param.marks >= 90}">Grade: A+</c:when>
      <c:when test="${param.marks >= 80}">Grade: A</c:when>
      <c:when test="${param.marks >= 70}">Grade: B</c:when>
      <c:when test="${param.marks >= 60}">Grade: C</c:when>
      <c:when test="${param.marks >= 35}">Grade: D</c:when>
      <c:otherwise>Grade: F (Fail)</c:otherwise>
    </c:choose>
  </c:if>
</body>
</html>
```

26. Create a JSP application using JSTL <c:forEach> to display a list of students in an HTML table.

students.jsp

```
<%@ page language="java" %>
<%@ page import="java.util.*" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<String[]> students = new ArrayList<>();

   students.add(new String[]{"1", "Dhruv", "MCA_01"});
   students.add(new String[]{"2", "Raj", "MCA"});
   students.add(new String[]{"3", "Nirav", "MCA"});

   request.setAttribute("students", students);
%>

<html>
<body>
   <h2>Student Records</h2>

   <table border="1" cellpadding="10">
      <tr>
         <th>ID</th>
         <th>Name</th>
         <th>Course</th>
      </tr>

      <c:forEach var="s" items="${students}">
         <tr>
            <td><c:out value="${s[0]}"/></td>
            <td><c:out value="${s[1]}"/></td>
            <td><c:out value="${s[2]}"/></td>
         </tr>
      </c:forEach>
   </table>
</body>
</html>
```

27. Create a JSP application using JSTL <c:forEach> to display employee records retrieved from a database.

employee.jsp

```
<%@ page language="java" %>
<%@ page import="java.sql.*" %>
<%@ page import="java.util.*" %>
```

```
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<Map<String, Object>> employees = new ArrayList<>();

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/javadb",
         "root",
         "1234"
      );

      Statement st = con.createStatement();
      ResultSet rs = st.executeQuery("SELECT * FROM employee");

      while (rs.next()) {
         Map<String, Object> emp = new HashMap<>();

         emp.put("id", rs.getInt("id"));
         emp.put("name", rs.getString("name"));
         emp.put("department", rs.getString("department"));
         emp.put("salary", rs.getDouble("salary"));

         employees.add(emp);
      }

      rs.close();
      st.close();
      con.close();

   } catch (Exception e) {
      request.setAttribute("error", e.getMessage());
   }

   request.setAttribute("employees", employees);
%>

<html>
<body>
   <h2>Employee Records</h2>

   <c:if test="${not empty error}">
      <p>Error: <c:out value="${error}"/></p>
   </c:if>

   <table border="1" cellpadding="10">
```

```
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Department</th>
        <th>Salary</th>
      </tr>

      <c:forEach var="e" items="${employees}">
        <tr>
          <td><c:out value="${e.id}"/></td>
          <td><c:out value="${e.name}"/></td>
          <td><c:out value="${e.department}"/></td>
          <td><c:out value="${e.salary}"/></td>
        </tr>
      </c:forEach>
    </table>
</body>
</html>
```

28. Create a JSP application demonstrating JSTL <c:set>, <c:remove> and <c:out>.

jstl.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<body>
    <h2>JSTL Core Tags</h2>

    <h3>c:set Example</h3>
    <c:set var="name" value="Dhruv"/>
    <c:set var="course" value="MCA_01"/>

    Name: <c:out value="${name}"/><br><br>
    Course: <c:out value="${course}"/><br><br>

    <h3>c:remove Example</h3>
    <c:remove var="name"/>

    Name after removal:
    <c:out value="${name}" default="Name Removed"/><br><br>

    <h3>c:out Example</h3>
```

```
   Roll No: <c:out value="01"/><br><br>
</body>
</html>
```

29. Create a JSP application using JSTL and EL to display product information stored in a collection.

Display:
Product ID
Product Name
Quantity
Price
Total Price

products.jsp

```
<%@ page language="java" %>
<%@ page import="java.util.*" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<Map<String, Object>> products = new ArrayList<>();

   Map<String, Object> p1 = new HashMap<>();
   p1.put("id", 101);
   p1.put("name", "Keyboard");
   p1.put("quantity", 2);
   p1.put("price", 500.0);
   products.add(p1);

   Map<String, Object> p2 = new HashMap<>();
   p2.put("id", 102);
   p2.put("name", "Mouse");
   p2.put("quantity", 3);
   p2.put("price", 250.0);
   products.add(p2);

   Map<String, Object> p3 = new HashMap<>();
   p3.put("id", 103);
   p3.put("name", "Monitor");
   p3.put("quantity", 1);
   p3.put("price", 5000.0);
   products.add(p3);
```

```
    request.setAttribute("products", products);
%>

<html>
<head>
    <title>Product Information</title>
</head>
<body>
    <h2>Product Information</h2>
    <h3>Name: Dhruv | Roll No: 01 | MCA_01</h3>

    <table border="1" cellpadding="10">
        <tr>
            <th>Product ID</th>
            <th>Product Name</th>
            <th>Quantity</th>
            <th>Price</th>
            <th>Total Price</th>
        </tr>

        <c:forEach var="p" items="${products}">
            <tr>
                <td><c:out value="${p.id}"/></td>
                <td><c:out value="${p.name}"/></td>
                <td><c:out value="${p.quantity}"/></td>
                <td><c:out value="${p.price}"/></td>
                <td>${p.quantity * p.price}</td>
            </tr>
        </c:forEach>
    </table>
</body>
</html>
```

30. Develop a Student Result Management System using JSP, JSTL and EL. The application should:
Accept student details and marks
Calculate total and percentage
Determine grade
Store student information in appropriate scope
Display the result using JSTL and EL
Avoid JSP scriptlets in the result page

input.jsp

```
<%@ page language="java" %>
<html>
<head>
   <title>Student Result System</title>
</head>
<body>
   <h2>Student Result Management System</h2>

   <form action="calculate.jsp" method="post">
      Student ID: <input type="text" name="id" required><br><br>
      Student Name: <input type="text" name="name" value="Dhruv" required><br><br>
      Roll No: <input type="text" name="rollno" value="01" required><br><br>
      Course: <input type="text" name="course" value="MCA_01" required><br><br>

      Subject 1: <input type="number" name="m1" min="0" max="100" required><br><br>
      Subject 2: <input type="number" name="m2" min="0" max="100" required><br><br>
      Subject 3: <input type="number" name="m3" min="0" max="100" required><br><br>
      Subject 4: <input type="number" name="m4" min="0" max="100" required><br><br>
      Subject 5: <input type="number" name="m5" min="0" max="100" required><br><br>

      <input type="submit" value="Calculate Result">
   </form>
</body>
</html>
```

calculate.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:set var="m1" value="${param.m1}" />
<c:set var="m2" value="${param.m2}" />
<c:set var="m3" value="${param.m3}" />
<c:set var="m4" value="${param.m4}" />
<c:set var="m5" value="${param.m5}" />

<c:set var="total" value="${m1 + m2 + m3 + m4 + m5}" />
<c:set var="percentage" value="${total / 5}" />

<c:set var="grade" value="F" />
<c:set var="status" value="Fail" />

<c:if test="${m1 >= 35 and m2 >= 35 and m3 >= 35 and m4 >= 35 and m5 >= 35}">
   <c:set var="status" value="Pass" />
```

```
    <c:choose>
        <c:when test="${percentage >= 90}">
            <c:set var="grade" value="A+" />
        </c:when>
        <c:when test="${percentage >= 80}">
            <c:set var="grade" value="A" />
        </c:when>
        <c:when test="${percentage >= 70}">
            <c:set var="grade" value="B" />
        </c:when>
        <c:when test="${percentage >= 60}">
            <c:set var="grade" value="C" />
        </c:when>
        <c:when test="${percentage >= 50}">
            <c:set var="grade" value="D" />
        </c:when>
        <c:otherwise>
            <c:set var="grade" value="E" />
        </c:otherwise>
    </c:choose>
</c:if>

<c:set var="studentId" value="${param.id}" scope="session" />
<c:set var="studentName" value="${param.name}" scope="session" />
<c:set var="rollno" value="${param.rollno}" scope="session" />
<c:set var="course" value="${param.course}" scope="session" />
<c:set var="total" value="${total}" scope="session" />
<c:set var="percentage" value="${percentage}" scope="session" />
<c:set var="grade" value="${grade}" scope="session" />
<c:set var="status" value="${status}" scope="session" />

<c:redirect url="result.jsp" />
```

result.jsp

```
<%@ page language="java" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<html>
<head>
    <title>Student Result</title>
</head>
<body>
    <h2>Student Result</h2>

    <table border="1" cellpadding="10">
        <tr>
```

```
        <th>Student ID</th>
        <td><c:out value="${sessionScope.studentId}" /></td>
      </tr>
      <tr>
        <th>Student Name</th>
        <td><c:out value="${sessionScope.studentName}" /></td>
      </tr>
      <tr>
        <th>Roll No</th>
        <td><c:out value="${sessionScope.rollno}" /></td>
      </tr>
      <tr>
        <th>Course</th>
        <td><c:out value="${sessionScope.course}" /></td>
      </tr>
      <tr>
        <th>Total Marks</th>
        <td><c:out value="${sessionScope.total}" /> / 500</td>
      </tr>
      <tr>
        <th>Percentage</th>
        <td><c:out value="${sessionScope.percentage}" />%</td>
      </tr>
      <tr>
        <th>Grade</th>
        <td><c:out value="${sessionScope.grade}" /></td>
      </tr>
      <tr>
        <th>Status</th>
        <td><c:out value="${sessionScope.status}" /></td>
      </tr>
    </table>

    <br>
    <a href="input.jsp">Back to Input Page</a>
</body>
</html>
```

31. Develop an Employee Management System using JSP, JSTL, EL and JDBC with the following operations:

Add Employee

Display Employee

Search Employee

Update Employee
Delete Employee

employee.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
<html>
<head>
  <title>Employee Management System</title>
</head>
<body>
  <h2>Employee Management System</h2>
  <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

  <h3>Add Employee</h3>
  <form action="EmployeeServlet" method="post">
    <input type="hidden" name="action" value="add">

    Employee ID: <input type="number" name="id" required><br><br>
    Name: <input type="text" name="name" required><br><br>
    Department: <input type="text" name="department" required><br><br>
    Salary: <input type="number" name="salary" step="0.01" min="0" required><br><br>

    <input type="submit" value="Add Employee">
  </form>

  <hr>

  <h3>Search Employee</h3>
  <form action="EmployeeServlet" method="get">
    <input type="hidden" name="action" value="search">
    Employee ID: <input type="number" name="id" required>
    <input type="submit" value="Search">
  </form>

  <hr>

  <h3>Update Employee</h3>
  <form action="EmployeeServlet" method="post">
    <input type="hidden" name="action" value="update">

    Employee ID: <input type="number" name="id" required><br><br>
    Name: <input type="text" name="name" required><br><br>
```

```
    Department: <input type="text" name="department" required><br><br>
    Salary: <input type="number" name="salary" step="0.01" min="0" required><br><br>

    <input type="submit" value="Update Employee">
</form>

<hr>

<h3>Delete Employee</h3>
<form action="EmployeeServlet" method="post">
    <input type="hidden" name="action" value="delete">
    Employee ID: <input type="number" name="id" required>
    <input type="submit" value="Delete Employee">
</form>

<hr>

<h3>Display All Employees</h3>
<form action="EmployeeServlet" method="get">
    <input type="hidden" name="action" value="display">
    <input type="submit" value="Display Employees">
</form>

<c:if test="${not empty message}">
    <h3><c:out value="${message}"/></h3>
</c:if>

<c:if test="${not empty employee}">
    <h3>Search Result</h3>
    <table border="1" cellpadding="8">
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
        </tr>
        <tr>
            <td><c:out value="${employee.id}"/></td>
            <td><c:out value="${employee.name}"/></td>
            <td><c:out value="${employee.department}"/></td>
            <td><c:out value="${employee.salary}"/></td>
        </tr>
    </table>
</c:if>

<c:if test="${not empty employees}">
    <h3>Employee List</h3>
    <table border="1" cellpadding="8">
```

```
        <tr>
          <th>ID</th>
          <th>Name</th>
          <th>Department</th>
          <th>Salary</th>
        </tr>

        <c:forEach var="e" items="${employees}">
          <tr>
            <td><c:out value="${e.id}"/></td>
            <td><c:out value="${e.name}"/></td>
            <td><c:out value="${e.department}"/></td>
            <td><c:out value="${e.salary}"/></td>
          </tr>
        </c:forEach>
      </table>
    </c:if>
</body>
</html>
```

Employee.java

```
package com.example;

public class Employee {
    private int id;
    private String name;
    private String department;
    private double salary;

    public Employee() {
    }

    public Employee(int id, String name, String department, double salary) {
        this.id = id;
        this.name = name;
        this.department = department;
        this.salary = salary;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
```

```
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getDepartment() {
        return department;
    }

    public void setDepartment(String department) {
        this.department = department;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }
}
```

EmployeeServlet.java

```
package com.example;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

@WebServlet("/EmployeeServlet")
public class EmployeeServlet extends HttpServlet {

    private static final String URL = "jdbc:mysql://localhost:3306/javadb";
    private static final String USER = "root";
```

```
private static final String PASSWORD = "1234";

private Connection getConnection() throws SQLException {
   return DriverManager.getConnection(URL, USER, PASSWORD);
}

@Override
public void init() throws ServletException {
   try {
      Class.forName("com.mysql.cj.jdbc.Driver");
   } catch (ClassNotFoundException e) {
      throw new ServletException(e);
   }
}

@Override
protected void doGet(HttpServletRequest request,
                HttpServletResponse response)
      throws ServletException, IOException {
   processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request,
                 HttpServletResponse response)
      throws ServletException, IOException {
   processRequest(request, response);
}

private void processRequest(HttpServletRequest request,
                       HttpServletResponse response)
      throws ServletException, IOException {

   response.setContentType("text/html;charset=UTF-8");
   request.setCharacterEncoding("UTF-8");

   String action = request.getParameter("action");

   try (Connection con = getConnection()) {

      if ("add".equals(action)) {
         String sql = "INSERT INTO employee (id, name, department, salary) VALUES (?, ?, ?, ?)";

         try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));
            ps.setString(2, request.getParameter("name"));
            ps.setString(3, request.getParameter("department"));
            ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));
```

```
            int rows = ps.executeUpdate();
            request.setAttribute("message",
                    rows > 0 ? "Employee added successfully!" : "Employee could not be added.");
        }

    } else if ("update".equals(action)) {
        String sql = "UPDATE employee SET name=?, department=?, salary=? WHERE id=?";

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setString(1, request.getParameter("name"));
            ps.setString(2, request.getParameter("department"));
            ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
            ps.setInt(4, Integer.parseInt(request.getParameter("id")));

            int rows = ps.executeUpdate();
            request.setAttribute("message",
                    rows > 0 ? "Employee updated successfully!" : "Employee not found.");
        }

    } else if ("delete".equals(action)) {
        String sql = "DELETE FROM employee WHERE id=?";

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));

            int rows = ps.executeUpdate();
            request.setAttribute("message",
                    rows > 0 ? "Employee deleted successfully!" : "Employee not found.");
        }

    } else if ("search".equals(action)) {
        String sql = "SELECT * FROM employee WHERE id=?";

        try (PreparedStatement ps = con.prepareStatement(sql)) {
            ps.setInt(1, Integer.parseInt(request.getParameter("id")));

            try (ResultSet rs = ps.executeQuery()) {
                if (rs.next()) {
                    Employee employee = new Employee(
                            rs.getInt("id"),
                            rs.getString("name"),
                            rs.getString("department"),
                            rs.getDouble("salary")
                    );
                    request.setAttribute("employee", employee);
                } else {
                    request.setAttribute("message", "Employee not found.");
```

```
                }
            }
        }

    } else if ("display".equals(action)) {
        List<Employee> employees = new ArrayList<>();
        String sql = "SELECT * FROM employee ORDER BY id";

        try (PreparedStatement ps = con.prepareStatement(sql);
             ResultSet rs = ps.executeQuery()) {

            while (rs.next()) {
                employees.add(new Employee(
                        rs.getInt("id"),
                        rs.getString("name"),
                        rs.getString("department"),
                        rs.getDouble("salary")
                ));
            }
        }

        request.setAttribute("employees", employees);
    }

} catch (NumberFormatException e) {
    request.setAttribute("message", "Please enter valid numeric values.");
} catch (SQLException e) {
    request.setAttribute("message", "Database error: " + e.getMessage());
}

request.getRequestDispatcher("employee.jsp").forward(request, response);
}
}
```

32. Develop a Product Management System using JSP, JSTL and EL. Retrieve product records from a database and display them using <c:forEach> and EL.

products.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="java.sql.*" %>
<%@ page import="java.util.*" %>
<%@ page import="com.example.Product" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>
```

```
<%
   List<Product> products = new ArrayList<>();

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/javadb",
         "root",
         "1234"
      );

      PreparedStatement ps = con.prepareStatement("SELECT * FROM product");
      ResultSet rs = ps.executeQuery();

      while (rs.next()) {
         products.add(new Product(
            rs.getInt("id"),
            rs.getString("name"),
            rs.getInt("quantity"),
            rs.getDouble("price")
         ));
      }

      rs.close();
      ps.close();
      con.close();

   } catch (Exception e) {
      request.setAttribute("error", e.getMessage());
   }

   request.setAttribute("products", products);
%>

<!DOCTYPE html>
<html>
<head>
   <title>Product Management System</title>
</head>
<body>
   <h2>Product Management System</h2>
   <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

   <c:if test="${not empty error}">
      <p><c:out value="${error}"/></p>
   </c:if>
```

```
    <table border="1" cellpadding="10">
        <tr>
            <th>Product ID</th>
            <th>Product Name</th>
            <th>Quantity</th>
            <th>Price</th>
            <th>Total Price</th>
        </tr>

        <c:forEach var="p" items="${products}">
            <tr>
                <td><c:out value="${p.id}"/></td>
                <td><c:out value="${p.name}"/></td>
                <td><c:out value="${p.quantity}"/></td>
                <td><c:out value="${p.price}"/></td>
                <td>${p.quantity * p.price}</td>
            </tr>
        </c:forEach>
    </table>
</body>
</html>
```

Product.java

```
package com.example;

public class Product {
    private int id;
    private String name;
    private int quantity;
    private double price;

    public Product(int id, String name, int quantity, double price) {
        this.id = id;
        this.name = name;
        this.quantity = quantity;
        this.price = price;
    }

    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }
```

```
    public int getQuantity() {
        return quantity;
    }

    public double getPrice() {
        return price;
    }
}
```

33. Create a JSP-based Login and Logout System using JSTL and EL. Use session scope to maintain the logged-in user and prevent access to the dashboard after logout.

login.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
</head>
<body>
    <h2>Login Page</h2>
    <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

    <c:if test="${not empty param.error}">
        <p>Invalid username or password!</p>
    </c:if>

    <form action="validate.jsp" method="post">
        Username:
        <input type="text" name="username" required>
        <br><br>

        Password:
        <input type="password" name="password" required>
        <br><br>

        <input type="submit" value="Login">
    </form>
</body>
</html>
```

validate.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
   <c:when test="${param.username == 'Dhruv' and param.password == '1234'}">
      <c:set var="loggedInUser" value="${param.username}" scope="session"/>
      <c:redirect url="dashboard.jsp"/>
   </c:when>

   <c:otherwise>
      <c:redirect url="login.jsp?error=1"/>
   </c:otherwise>
</c:choose>
```

dashboard.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
   <c:when test="${empty sessionScope.loggedInUser}">
      <c:redirect url="login.jsp"/>
   </c:when>

   <c:otherwise>
      <!DOCTYPE html>
      <html>
      <head>
         <title>Dashboard</title>
      </head>
      <body>
         <h2>Welcome, <c:out value="${sessionScope.loggedInUser}"/>!</h2>
         <p>You are successfully logged in.</p>

         <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

         <form action="logout.jsp" method="post">
            <input type="submit" value="Logout">
         </form>
      </body>
      </html>
   </c:otherwise>
```

```
</c:choose>
```

logout.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:remove var="loggedInUser" scope="session"/>

<%
   session.invalidate();
   response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate");
   response.setHeader("Pragma", "no-cache");
   response.setDateHeader("Expires", 0);
%>

<c:redirect url="login.jsp"/>
```

34. Develop a JSP application that retrieves records from a database and uses JSTL <c:forEach>, <c:if> and EL to display records conditionally.

students.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="java.sql.*" %>
<%@ page import="java.util.*" %>
<%@ page import="com.example.Student" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
   List<Student> students = new ArrayList<>();

   try {
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/javadb",
         "root",
         "1234"
      );

      PreparedStatement ps = con.prepareStatement("SELECT * FROM student");
      ResultSet rs = ps.executeQuery();
```

```
        while (rs.next()) {
            students.add(new Student(
                rs.getInt("id"),
                rs.getString("name"),
                rs.getString("course"),
                rs.getInt("semester"),
                rs.getDouble("marks")
            ));
        }

        rs.close();
        ps.close();
        con.close();

    } catch (Exception e) {
        request.setAttribute("error", e.getMessage());
    }

    request.setAttribute("students", students);
%>

<!DOCTYPE html>
<html>
<head>
    <title>Student Records</title>
</head>
<body>
    <h2>Student Records</h2>
    <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

    <c:if test="${not empty error}">
        <p>Database Error: <c:out value="${error}"/></p>
    </c:if>

    <c:if test="${empty students and empty error}">
        <p>No student records found.</p>
    </c:if>

    <c:if test="${not empty students}">
        <table border="1" cellpadding="10">
            <tr>
                <th>ID</th>
                <th>Name</th>
                <th>Course</th>
                <th>Semester</th>
                <th>Marks</th>
                <th>Result</th>
            </tr>
```

```
        <c:forEach var="s" items="${students}">
          <tr>
            <td><c:out value="${s.id}"/></td>
            <td><c:out value="${s.name}"/></td>
            <td><c:out value="${s.course}"/></td>
            <td><c:out value="${s.semester}"/></td>
            <td><c:out value="${s.marks}"/></td>

            <td>
              <c:if test="${s.marks >= 40}">
                Pass
              </c:if>
              <c:if test="${s.marks < 40}">
                Fail
              </c:if>
            </td>
          </tr>
        </c:forEach>
      </table>
    </c:if>
</body>
</html>
```

Student.java

```
package com.example;

public class Student {
    private int id;
    private String name;
    private String course;
    private int semester;
    private double marks;

    public Student(int id, String name, String course, int semester, double marks) {
        this.id = id;
        this.name = name;
        this.course = course;
        this.semester = semester;
        this.marks = marks;
    }

    public int getId() {
        return id;
    }
```

```
    public String getName() {
        return name;
    }

    public String getCourse() {
        return course;
    }

    public int getSemester() {
        return semester;
    }

    public double getMarks() {
        return marks;
    }
}
```

35. Develop a College Student Management System using JSP, JSTL, EL and JDBC. The system should provide:

Student Registration
Student Login
Student List
Search Student
Update Student
Delete Student
Session Management
Logout

Student.java

```
package com.example;

public class Student {
    private int id;
    private String name;
    private String email;
    private String course;
    private String password;

    public Student() {}

    public Student(int id, String name, String email, String course, String password) {
        this.id = id;
```

```
        this.name = name;
        this.email = email;
        this.course = course;
        this.password = password;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }

    public String getCourse() { return course; }
    public void setCourse(String course) { this.course = course; }

    public String getPassword() { return password; }
    public void setPassword(String password) { this.password = password; }
}
```

DBConnection.java

```
package com.example;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/javadb";
    private static final String USER = "root";
    private static final String PASSWORD = "1234";

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

register.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
```

```
<html>
<head><title>Student Registration</title></head>
<body>
  <h2>College Student Registration</h2>
  <p>Name: Dhruv | Roll No: 01 | MCA_01</p>

  <c:if test="${not empty param.error}">
    <p>Registration failed. Email may already exist.</p>
  </c:if>

  <form action="StudentServlet" method="post">
    <input type="hidden" name="action" value="register">

    Student ID: <input type="number" name="id" required><br><br>
    Name: <input type="text" name="name" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    Course: <input type="text" name="course" value="MCA_01" required><br><br>
    Password: <input type="password" name="password" required><br><br>

    <input type="submit" value="Register">
  </form>

  <a href="login.jsp">Login</a>
</body>
</html>
```

login.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<!DOCTYPE html>
<html>
<head><title>Student Login</title></head>
<body>
  <h2>Student Login</h2>

  <c:if test="${not empty param.error}">
    <p>Invalid email or password.</p>
  </c:if>

  <form action="StudentServlet" method="post">
    <input type="hidden" name="action" value="login">

    Email: <input type="email" name="email" required><br><br>
    Password: <input type="password" name="password" required><br><br>
```

```
        <input type="submit" value="Login">
    </form>

    <a href="register.jsp">Register New Student</a>
</body>
</html>
```

dashboard.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
    <c:when test="${empty sessionScope.studentId}">
        <c:redirect url="login.jsp"/>
    </c:when>
    <c:otherwise>
        <!DOCTYPE html>
        <html>
        <head><title>Student Dashboard</title></head>
        <body>
            <h2>Welcome, <c:out value="${sessionScope.studentName}"/>!</h2>

            <p>Student ID: <c:out value="${sessionScope.studentId}"/></p>
            <p>Email: <c:out value="${sessionScope.studentEmail}"/></p>
            <p>Course: <c:out value="${sessionScope.studentCourse}"/></p>

            <a href="StudentServlet?action=list">Student List</a><br><br>
            <a href="search.jsp">Search Student</a><br><br>
            <a href="logout.jsp">Logout</a>
        </body>
        </html>
    </c:otherwise>
</c:choose>
```

search.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:if test="${empty sessionScope.studentId}">
    <c:redirect url="login.jsp"/>
</c:if>

<!DOCTYPE html>
```

```
<html>
<head><title>Search Student</title></head>
<body>
  <h2>Search Student</h2>

  <form action="StudentServlet" method="get">
    <input type="hidden" name="action" value="search">
    Enter Student ID:
    <input type="number" name="id" required>
    <input type="submit" value="Search">
  </form>

  <c:if test="${not empty param.error}">
    <p>Student not found.</p>
  </c:if>

  <a href="dashboard.jsp">Back to Dashboard</a>
</body>
</html>
```

students.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:if test="${empty sessionScope.studentId}">
  <c:redirect url="login.jsp"/>
</c:if>

<!DOCTYPE html>
<html>
<head><title>Student List</title></head>
<body>
  <h2>College Student List</h2>

  <c:if test="${empty students}">
    <p>No students found.</p>
  </c:if>

  <c:if test="${not empty students}">
    <table border="1" cellpadding="8">
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Email</th>
        <th>Course</th>
        <th>Actions</th>
```

```
        </tr>

        <c:forEach var="s" items="${students}">
          <tr>
            <td><c:out value="${s.id}"/></td>
            <td><c:out value="${s.name}"/></td>
            <td><c:out value="${s.email}"/></td>
            <td><c:out value="${s.course}"/></td>
            <td>
              <a
href="edit.jsp?id=s.id&name =${s.name}&email=s.email&course =${s.course}">Edit</a>
              |
              <form action="StudentServlet" method="post" style="display:inline">
                <input type="hidden" name="action" value="delete">
                <input type="hidden" name="id" value="${s.id}">
                <input type="submit" value="Delete"
                       onclick="return confirm('Delete this student?')">
              </form>
            </td>
          </tr>
        </c:forEach>
      </table>
    </c:if>

    <br><a href="dashboard.jsp">Back to Dashboard</a>
</body>
</html>
```

edit.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:if test="${empty sessionScope.studentId}">
    <c:redirect url="login.jsp"/>
</c:if>

<!DOCTYPE html>
<html>
<head><title>Update Student</title></head>
<body>
    <h2>Update Student</h2>

    <form action="StudentServlet" method="post">
        <input type="hidden" name="action" value="update">

        Student ID:
```

```
        <input type="number" name="id" value="<c:out value='${param.id}'/>" readonly><br><br>

        Name:
        <input type="text" name="name" value="<c:out value='${param.name}'/>" required><br><br>

        Email:
        <input type="email" name="email" value="<c:out value='${param.email}'/>"
required><br><br>

        Course:
        <input type="text" name="course" value="<c:out value='${param.course}'/>"
required><br><br>

        <input type="submit" value="Update Student">
    </form>
</body>
</html>
```

logout.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<%
    session.invalidate();
    response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate");
    response.setHeader("Pragma", "no-cache");
    response.setDateHeader("Expires", 0);
%>

<c:redirect url="login.jsp"/>
```

StudentServlet.java

```
package com.example;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

import java.io.IOException;
import java.sql.*;
import java.util.ArrayList;
```

```
import java.util.List;

@WebServlet("/StudentServlet")
public class StudentServlet extends HttpServlet {

    @Override
    public void init() throws ServletException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            throw new ServletException(e);
        }
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    private void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        request.setCharacterEncoding("UTF-8");
        response.setContentType("text/html;charset=UTF-8");

        String action = request.getParameter("action");

        try (Connection con = DBConnection.getConnection()) {

            if ("register".equals(action)) {
                String sql = "INSERT INTO student (id, name, email, course, password) VALUES (?, ?, ?,
?, ?)";

                try (PreparedStatement ps = con.prepareStatement(sql)) {
                    ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                    ps.setString(2, request.getParameter("name"));
                    ps.setString(3, request.getParameter("email"));
                    ps.setString(4, request.getParameter("course"));
                    ps.setString(5, request.getParameter("password"));
                    ps.executeUpdate();
                }
```

```
    response.sendRedirect("login.jsp");
    return;

} else if ("login".equals(action)) {
    String sql = "SELECT * FROM student WHERE email=? AND password=?";

    try (PreparedStatement ps = con.prepareStatement(sql)) {
        ps.setString(1, request.getParameter("email"));
        ps.setString(2, request.getParameter("password"));

        try (ResultSet rs = ps.executeQuery()) {
            if (rs.next()) {
                HttpSession session = request.getSession();
                session.setAttribute("studentId", rs.getInt("id"));
                session.setAttribute("studentName", rs.getString("name"));
                session.setAttribute("studentEmail", rs.getString("email"));
                session.setAttribute("studentCourse", rs.getString("course"));

                response.sendRedirect("dashboard.jsp");
            } else {
                response.sendRedirect("login.jsp?error=1");
            }
        }
    }
    return;

} else {
    HttpSession session = request.getSession(false);
    if (session == null || session.getAttribute("studentId") == null) {
        response.sendRedirect("login.jsp");
        return;
    }
}

if ("list".equals(action)) {
    List<Student> students = new ArrayList<>();

    try (PreparedStatement ps =
             con.prepareStatement("SELECT id, name, email, course FROM student");
         ResultSet rs = ps.executeQuery()) {

        while (rs.next()) {
            students.add(new Student(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("email"),
                    rs.getString("course"),
```

```
                ""
            ));
        }
    }

    request.setAttribute("students", students);
    request.getRequestDispatcher("students.jsp").forward(request, response);
    return;
}

if ("search".equals(action)) {
    try (PreparedStatement ps = con.prepareStatement(
            "SELECT id, name, email, course FROM student WHERE id=?")) {

        ps.setInt(1, Integer.parseInt(request.getParameter("id")));

        try (ResultSet rs = ps.executeQuery()) {
            if (rs.next()) {
                Student s = new Student(
                        rs.getInt("id"),
                        rs.getString("name"),
                        rs.getString("email"),
                        rs.getString("course"),
                        ""
                );
                request.setAttribute("students", List.of(s));
                request.getRequestDispatcher("students.jsp").forward(request, response);
            } else {
                response.sendRedirect("search.jsp?error=1");
            }
        }
    }
    return;
}

if ("update".equals(action)) {
    String sql = "UPDATE student SET name=?, email=?, course=? WHERE id=?";

    try (PreparedStatement ps = con.prepareStatement(sql)) {
        ps.setString(1, request.getParameter("name"));
        ps.setString(2, request.getParameter("email"));
        ps.setString(3, request.getParameter("course"));
        ps.setInt(4, Integer.parseInt(request.getParameter("id")));
        ps.executeUpdate();
    }

    response.sendRedirect("StudentServlet?action=list");
    return;
```

```
            }

            if ("delete".equals(action)) {
                try (PreparedStatement ps =
                        con.prepareStatement("DELETE FROM student WHERE id=?")) {
                    ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                    ps.executeUpdate();
                }

                response.sendRedirect("StudentServlet?action=list");
                return;
            }

            response.sendRedirect("dashboard.jsp");

        } catch (SQLException | NumberFormatException e) {
            throw new ServletException("Student management operation failed.", e);
        }
    }
}
```